Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Sai Thotakura	tsai@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

I like exercising and going to the gym so my first interest was in making a workout tracker with Drafter that allows the user to do multiple inputs that help them track their workout and progress. I also want the ability to create a workout plan for each day of the week. Another goal I have is to include a progression tracker using plots to show users a visual representation of their progress on a specific exercise.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

On the Apple App store, I found many workout tracker apps like Hevy, Liftoff, Setgraph, and more. They all have workout tracking capabilities and some of them allow you to specifically log sets for each exercise, like recording what weight you were able to lift for a specific number of reps on that particular day. These all have very flashy AI and I've noticed social media ads about some of them too, so they seem to attract many users that way.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I would like this to be an easy to use web application that allows users to set up custom workout plans or potentially select a workout preset from some predetermined workout plans. I also want the user to be able to log the exact weight and reps of a specific exercise they did in a single day.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

My program will take input in the form of the input function that will allow the user to enter a weight (in lbs), number of sets, and reps per set. These will be in the form of integers. To process the inputs, my program will calculate a total volume per exercise in that workout and save it to a list so the user can then see a plot of their workout progression (hopefully a positive trend over time). I'll do this using what we've learned on the chapter assignments with matplotlib.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will need to input strings to enter in the name of the workouts. Then, once they start a workout, they need to enter integer values that represent their number of sets of a particular exercise/movement, weight, and repetitions. I was also thinking to require physical attributes of the user like height/weight but I feel that isn't useful for this app.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

An easy way to process the raw data (sets, reps, weight) is to just multiply all of them and then save that number to a list. Then later on, if the user wants to see how they're progressing with certain workouts, they can view the progression for a specific workout by inputting that string. I'm thinking that whatever they input will trigger the program to display the plot showing the progression for that specific exercise.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will get back a plot showing their progression in a specific exercise. But before then, I think it'd be cool if when a user increases weight or does more repetitions than the previous time they performed that exercise, there could be some sort of congratulation message.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Workout, Tracker, Exercise, Progress

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Exercise: name, muscle group, number of sets, number of reps *Ignore: rest time, gym location, time of day, heart rate*

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

Saving the data from every exercise into a list so that the user can see a plotted version sounds very difficult to implement for me. I'm hoping that an LLM will simplify the process and return satisfying results. Also I feel like there's a lot of possibility as far as workouts and exercises go, so I need a way to ensure I can organize all of the workouts that a person may

do, especially if they do many sets of an exercise or a large variety.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will see the outputted code that the LLM gives me and read through each line to see any familiarities with what we've learned so far in class. If I see something I understand but is written in a different manner, I'll attempt to rewrite it so it fits my understanding better. But if I see code that I've never seen before, I'll ask the LLM to expand on it or simplify it so it's written in a way I understand. Also, instead of copying and pasting, I'll type out the code so I see each function and line instead of just blindly following the LLM's instructions.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My goal is to build a workout tracker program that actually works and has minimal to no bugs. I want it to be simple to use and intuitive enough because I know some apps out there are filled with confusing UI and a hundred buttons redirecting to useless features. Also, I hope to learn how to effectively use AI to code. Although I have no interest in pursuing Computer Science, my dad who's a software engineer told me that at his company, all employees are expected to use AI to code now, so it's obviously a useful skillset to have.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to develop a Drafter program website that allows users to track their workouts and

see their progression over time. What are 3 common problems students face when tracking workouts, and what are 3 simple features this tool should have?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

The LLM told me some problems people face when working out like forgetting weights/reps of previous workouts, identifying if they're actually progressing across their exercises, and also becoming demotivated in tracking every workout. To solve this, the AI suggested a basic workout logger common throughout many apps already on the App store. This would allow the user to select an exercise and then enter the weight, sets, and reps of that particular exercise on that day. The AI also suggested a progress page which shows a plot representing the progression in a particular exercise over time. This already exists in many apps for a good reason because it lets you view how much you've improved after working out consistently. Finally, the AI suggested a history tracker that lets the user see what exact workouts they did previously.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

The Progress Dashboard seems very complicated. I think we'd have to save the workout numbers inputted by the user to a list and then when the user tries to access a plot of a specific workout, it would need to use matplotlib or something similar to create a plot of the progression. Are there any simpler alternatives?
LLM answer summary: It suggested other alternatives but the most relevant one that I liked was just a simple table representing the data. The table had 3 columns: Date, Exercise, Reps & Sets. This technique showed the data in a very simple way without having to use plots or other advanced methods. It works by storing workouts into a record and then filtering them by exercise name and finally displaying the data on a clean table.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

After this conversation with AI, my direction still remains largely the same. However, I think I'll have to rethink some of the features I was thinking about because they may be difficult to implement with my current coding skill. Most of the things that the LLM told me were ideas I was already thinking about so in a way, the conversation reassured me in knowing I have the ability to carry on with this project idea.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I will use an LLM to write more complicated code and if I don't understand a specific part of it, I'll ask the LLM to explain more about the code in depth. Also, I'll refer to the BlockPy Textbook for some ideas we've learned in class if I need a brush up. I think right now, AI provides the fastest and most useful way to seek help for coding issues.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I will identify what lines the errors are coming from and first see if there's any basic errors I may have overlooked. Usually the console says what type of error it is so that could make the troubleshooting process very simple. If I can't figure it out, I'll paste the issue into the LLM and see what it thinks of the issue. Most likely, it will find the error and send an immediate fix, but rather than just pasting the new code segment in, I'll make the effort to understand why the code went wrong.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I only used an LLM for Part 5. Here's my conversation with ChatGPT:

I want to develop a Drafter program website that allows users to track their workouts and see their progression over time. What are 3 common problems students face when tracking workouts, and what are 3 simple features this tool should have?
For a college student-focused workout tracker, I would focus on making workout logging quick, keeping track of progress, and helping students stay consistent. A lot of students don't need an overly complicated fitness app; they just want an easy way to record workouts and see whether they're improving.

1. Three common problems students face
Problem

Why it's frustrating

1. Forgetting previous workouts

Students forget how much weight they lifted, how many reps they completed, or which exercises they performed last time.

2. Difficulty tracking progression

It's hard to tell whether strength is improving when workout information is scattered across notes, apps, or memory.

3. Inconsistent workout logging

Logging every exercise, set, and rep can be time-consuming, so students sometimes stop recording their workouts altogether.

2. Three simple features your website should have
CURVA - Free Workout App for Team Sport Players | Track Lifts & Training Plans

1. Workout Logger

Allow users to select exercises and enter the weight, sets, and reps they completed.

Add exercises to a workout.

Record weight and reps.

Save completed workouts.

'It’s no longer enough for an app to tell you what to do. People want to know why': Fitness app Fitbod's founder on the reason behind the AI fitness boom | TechRadar

2. Progress Dashboard

Display how users are progressing with individual exercises over time.

View previous weights and reps.

Show a graph of weight progression.

Display personal bests.

Fitness UI UX | Home Onboarding by Sadax Studio on Dribbble

3. Workout History

Let users look back at their previous workouts and see how consistently they've been exercising.

View workouts by date.

See which exercises were completed.

Track weekly workout frequency.

3. How to organize the Drafter website
I'd keep the website to just three main pages:

Dashboard: Quick overview of recent workouts, weekly activity, and personal bests.

Log Workout: Enter exercises, sets, reps, and weights.

Progress: Graphs and workout history for tracking improvement.

My recommendation: Make the Workout Logger the main focus. The biggest challenge is getting students to consistently record their workouts. If logging takes only a few seconds per exercise, they're more likely to use the website.

This is also a manageable project for Drafter because you can build it without needing complicated algorithms, external APIs, or AI recommendations.

I can help turn these features into a simple Drafter page-by-page implementation next.

Develop workout tracker in Drafter

The Progress Dashboard seems very complicated. I think we'd have to save the workout numbers inputted by the user to a list and then when the user tries to access a plot of a specific workout, it would need to use matplotlib or something similar to create a plot of the progression. Are there any simpler alternatives?
Yes. I would actually simplify the Progress Dashboard significantly. You don't need Matplotlib, graphs, or complicated data processing to make workout progression useful.

Here are three alternatives, ordered from easiest to more advanced.

1. Personal best tracker (easiest)

Instead of plotting progression, simply display the user's best recorded performance for each exercise.

My Personal Bests

Bench Press

Best weight recorded

135 lbs
Lat Pulldown

Best weight recorded

120 lbs
Bicep Curl

Best weight recorded

30 lbs
How it works in Python:

Store the highest weight entered for each exercise.

When a user logs a new workout, compare the new weight with their existing record.

Update the personal best if the new weight is higher.

No graphs required!

2. Previous workout comparison
Show the user's most recent performance alongside their previous performance for a selected exercise.

Bench Press

Compare your last two workouts

Previous workout

95 lbs
2 sets × 5 reps

Most recent

100 lbs

2 sets × 5 reps

+5 lbs improvement
How it works in Python:

Save workout entries in a list.

Find the two most recent entries for the selected exercise.

Display their weights and reps.

Subtract the previous weight from the current weight.

This only requires basic lists, loops, and conditionals.

3. Exercise history table
Rather than generating graphs, display a simple table containing all previous performances.

Bench Press History

Date

Weight

Sets × Reps

Oct 5

85 lbs

2 × 5

Oct 12

90 lbs

2 × 5

Oct 19

95 lbs

2 × 5

Oct 26

100 lbs

2 × 5

Example workout records

How it works in Python:

Store each workout as a record.

Filter records by exercise name.

Display matching records in a table.

My recommendation for your Drafter project
I'd combine features 1 and 2.

Feature

Difficulty

Purpose

Personal best tracker

Easy

Shows users their best performances

Previous workout comparison

Easy

Shows whether they're improving

Workout logger

Easy

Records sets, reps, and weights

You could organize your website into just two pages:

Workout Logger: Enter exercises, weights, sets, and reps.

My Progress: View personal bests and compare your latest workout with your previous one.

One important programming consideration: You will still need to store workout information somewhere, such as a list of records. However, you don't need to use Matplotlib or any external graphing library. The calculations can be handled with basic Python.

This would make a much more manageable Drafter project while still solving the original problem of students struggling to track their gym progress.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.